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ChatGPT Research: A Bibliometric Analysis Based on the Web of Science from 2023 to June 2024

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Abstract: ChatGPT, or Chat Generative Pre-trained Transformer, developed by OpenAI, is a versatile chatbot known for generating human-like text responses. Since its launch in November 2022, it has sparked interest and debate. This bibliometric study aimed to explore ChatGPT-related publications using the Web of Science database from 2023 to June 2024. Original articles in English were retrieved on 24 June 2024, using the topic field "ChatGPT". Citation records were analyzed using bibliometrix 4.1 and VOSviewer 1.6.20. Between January 2023 and 24 June 2024, 3231 original articles on ChatGPT were published in 1404 journals, with an average citation rate of 5.6 per article. The United States led with 877 articles, followed by China and India. The University of California System, Harvard University, and the State University System of Florida were the most prolific institutions. Keyword co-occurrence network analysis revealed the interdisciplinary nature of ChatGPT research, particularly contributions in healthcare, education, and technology. In conclusion, this bibliometric analysis identified critical areas of ChatGPT research focus, such as applications in educational settings and its ethical implications. These findings are crucial for fostering further advancements that leverage ChatGPT's capabilities while mitigating its risks.

Keywords: ChatGPT; generative artificial intelligence; bibliometrix; VOSviewer

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1. Introduction

ChatGPT, short for Chat Generative Pre-trained Transformer, is an innovative artificial intelligence (AI) language model-based chatbot developed by OpenAI (OpenAI, L.L.C., San Francisco, CA, USA). Its underlying transformer architecture enables it to understand and generate human-like text responses within a conversational context, making it a versatile tool for various applications [1]. Since its official debut in November 2022, ChatGPT-3.5 has attracted considerable attention from both the general public and the scholarly community [2]. In academia, ChatGPT's arrival has sparked a wave of research and exploration into its potential applications across numerous disciplines, especially in the fields of healthcare [3,4] and education [5,6]. The debut of GPT-4 in February 2023 marked further improvements in its accuracy and functionality, including the ability to process to images [7,8].

Given the considerable quantity and diverse range of publications dedicated to ChatGPT in recent years, comprehending the full scholarly output on this subject can be challenging. Bibliometrics, a quantitative method for analyzing the scholarly literature, can be instrumental in navigating this complexity by providing systematic and quantitative analyses of the literature. This approach enables researchers to identify influential authors, journals, and institutions, and emerging areas of interest through metrics such as publication counts, citation analysis, and co-authorship networks [9].

Several studies on ChatGPT using bibliometric methods have been published. For instance, Farhat et al. conducted a bibliometric analysis on 533 articles using the Scopus database to explore ChatGPT's early footprint in academia [10]. Alessandri-Bonetti et al. examined 724 articles retrieved from PubMed using the keyword "ChatGPT" up to 1 July 2023 [11]. Khosravi et al. identified 2531 and 5839 documents related to chatbot-focused research from the Web of Science and Scopus databases, respectively. A bibliometric analysis, including keyword co-occurrence analysis on 211 publications, was conducted [12]. Baber et al. conducted a bibliometric analysis of 328 research articles on ChatGPT extracted from Scopus database [13]. Ahmed and Sab examined 333 ChatGPT indexed in the Web of Science up to June 2023 [14]. Liu et al. explored global trends and hotspots of ChatGPT based on 1239 publications indexed by the Web of Science from January 2023 to January 2024 [15].

There were also several studies focused on health- and medicine-related publications. Wu et al. examined the bibliometric properties of 247 articles related to ChatGPT, medicine, and nursing based on six electronic databases, including PubMed, CINAHL, Embase, Web of Science, IEEE Xplore, and ACM Digital Library from November 2022 to August 2023 [16]. Li et al. analyzed 574 articles related to applications of ChatGPT in medicine, covering the period from January 2000 up to January 2024 [17]. Gande et al. explored 786 articles using the keyword "ChatGPT" published in medical-related journals from January to September 2023 [18]. Moreover, Barrington et al. focused on 267 primary studies in the literature on ChatGPT in the medical field using PubMed, Embase, Scopus, and Web of Science databases from January 2023 to July 2023 [19]. Furthermore, several studies have used bibliometric methods to explore ChatGPT in specific disciplines, such as obstetrics and gynecology [20], educational research [21], social sciences [22], plastic surgery [23], nursing education [24], and anesthesiology [25].

Despite the insights provided by previous bibliometric studies, the ongoing advances in ChatGPT technology demand continuous and timely updates to these analyses. As the field of ChatGPT-related research evolves rapidly, maintaining the relevance and accuracy of bibliometric data are crucial. This study aimed to provide an up-to-date bibliometric analysis, reflecting the latest developments up to June 2024, using data from the Web of Science database. This study also incorporated thematic mapping to identify and visualize key research themes and trends in ChatGPT research.

2. Materials and Methods

2.1. Source of Bibliometric Data and Search Query

The data for this bibliometric study were retrieved from the Science Citation Index Expanded™ (SCI-Expanded™) in the Web of Science Core Collection (Clarivate Analytics, Philadelphia, PA, USA). A study flow diagram, constructed with reference to the proposed reporting guidelines for bibliometric studies, is shown in Figure 1 [26].

The Web of Science database was selected for this study for two primary reasons. First, the Web of Science Journal Impact Factor (JIF) is a widely recognized and accepted citation-based metric for assessing journal quality [27]. Second, the availability of comprehensive citation data in the Web of Science enables the detailed analysis and visualization of relationships between articles.

The search was conducted on a single day, 24 June 2024. The SCI-Expanded in the Web of Science Core Collection was queried using the Topic (TS) field: TS = (chatgpt) OR TS = (chat-gpt) OR TS = (chat gpt). To ensure accurate interpretation of the results, the publication language was restricted to English, as indicated by the Web of Science field tag LA.

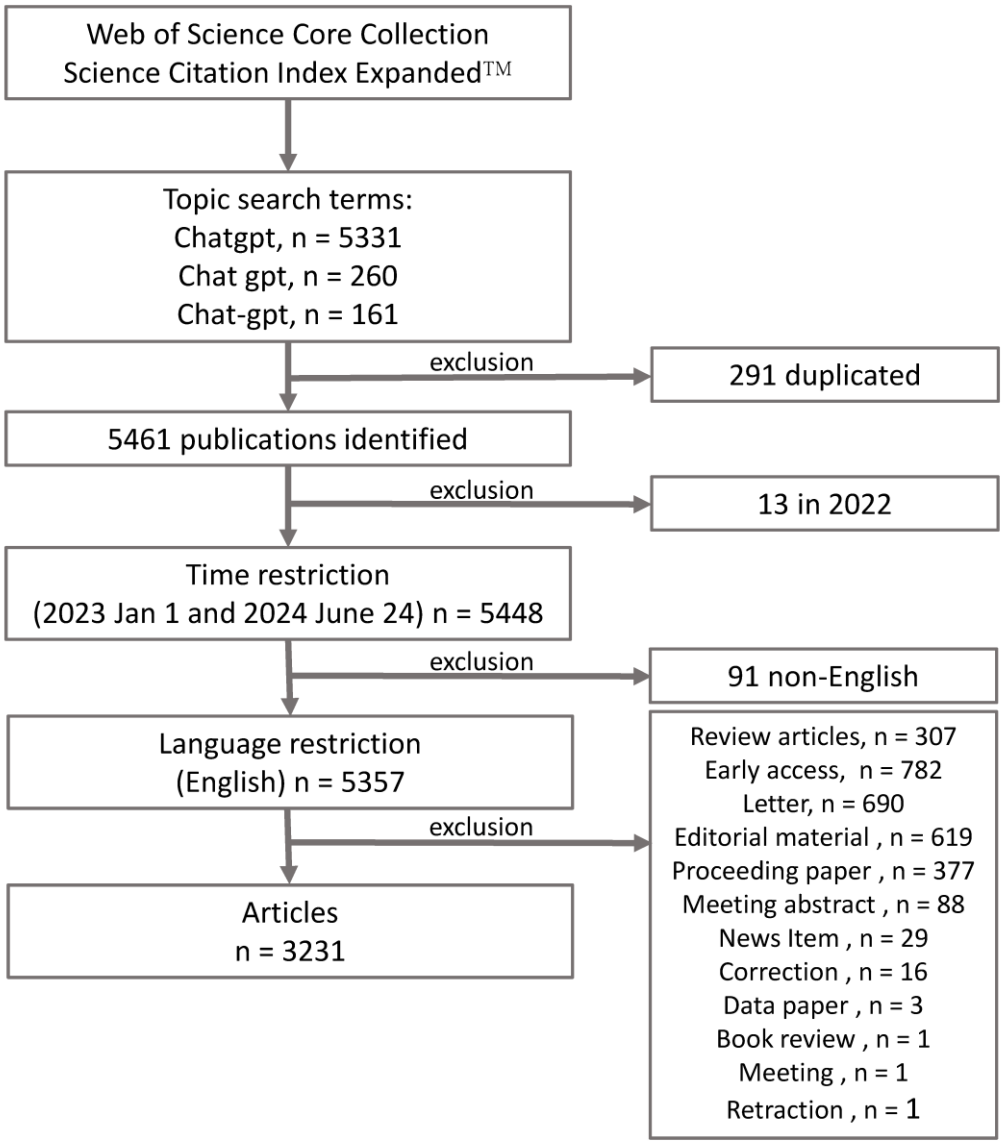


Figure 1. Study flow diagram.

2.2. Bibliometric Analysis

The complete citation records obtained from the search was imported into the bibliometric software bibliometrix 4.1 (Naples, Italy) [28]. The shiny app Biblioshiny was used to provide a graphical web interface in the R environment (version R-4.4.2) using the RStudio interface (version 2024.04.2). Analyses of author performance, publication analysis (by countries and institutions performance), and journal impact were conducted. Furthermore, a thematic mapping analysis was performed visually based on Callon's centrality and density ranking [29]. The thematic map displays network cluster centrality on the X-axis, indicating the importance of research themes. The Y-axis measures density, which reflects the internal robustness and development potential of a cluster network [30]. The map is divided into four quadrants, each representing different types of themes: (a) motor themes in the first quadrant (top right), where the cluster network exhibits both high centrality and density, indicating that these themes are well-established and pivotal in ChatGPT research; (b) niche themes in the second quadrant (top left), characterized by high density but low centrality, suggesting that they are well-developed but not broadly influential; (c) emerging or declining themes in the third quadrant (bottom left), marked by low centrality and density, implying that they are minimally developed and marginal; and (d) basic themes

in the fourth quadrant (bottom right) are characterized by high centrality but low density, indicating that they are important for transdisciplinary issues but not yet fully explored.

VOSviewer 1.6.20 for Microsoft Windows (Centre for Science and Technology Studies, Leiden University, The Netherlands) [31] was used to construct and visualize keyword co-occurrence network (author-supplied keywords and "Keywords Plus") to identify clusters of related terms and their frequency of occurrence. "Keywords Plus" is a feature in the Web of Science that enhances the traditional keyword-based search method by automatically adding relevant terms extracted from the titles of articles cited in the references of a given article. These terms are not necessarily present in the title, abstract, or author keywords of the article itself but are deemed important based on their appearance in the titles of cited works. This approach aims to capture additional relevant topics that the original keywords might have missed.

3. Results

3.1. Overview and Trends

This bibliometric analysis covered an 18-month period from January 2023 to June 2024. During this time, 3231 original articles were analyzed, and they were published in 1404 journals. The annual growth rate of publications was 37.6%. On average, each article received 5.6 citations and contributed to a total of 88,158 references. This study involved a total of 12,680 authors. Among these, 449 authors contributed single-authored articles, indicating that while single-author contributions were present, most of the research was collaborative. In addition, 26.7% of the articles involved international co-authorship.

3.2. Distribution of Countries

Table 1 presents the top 10 productive countries based on the corresponding author's affiliation in ChatGPT research indexed by the Web of Science from January 2023 to June 2024. The United States of America led with 877 articles, accounting for 27.1% of the total. Of these, 741 were single-country publications (SCP), and 136 were multiple-country publications (MCP), with an MCP percentage of 15.5%. China ranked second with 416 articles (12.9%), including 292 SCP and 124 MCP, resulting in a higher MCP percentage of 29.8%. Moreover, India and Germany each contributed 141 articles (4.4%). India's publications included 108 SCP and 33 MCP, with an MCP percentage of 23.4%. Germany had 99 SCP and 42 MCP, matching China's MCP percentage of 29.8%. Overall, the mean MCP percentage across these top 10 productive countries was 38.1%, showing substantial international collaboration in ChatGPT research.

Table 1. Top 10 productive countries based on corresponding author's affiliation in ChatGPT research indexed by the Web of Science from January 2023 to June 2024.

Rank	Countries	Number of Articles (%)	Single Country Publication	Multiple Country Publication	Multiple Country Publication %
1	United States of America	877 (27.1)	741	136	15.5
2	China	416 (12.9)	292	124	29.8
3	India	141 (4.4)	108	33	23.4
4	Germany	141 (4.4)	99	42	29.8
5	United Kingdom	137 (4.2)	82	55	40.1
6	Australia	102 (3.2)	78	24	23.5
7	South Korea	93 (2.9)	71	22	23.7
8	Italy	77 (2.4)	40	37	48.1
9	Canada	69 (2.1)	41	28	40.6
10	Spain	68 (2.1)	51	17	25.0
All	Mean	–	–	–	38.1

3.3. Distribution of Research Institutions

Table 2 shows the top 10 productive institutions in ChatGPT research indexed by the Web of Science from January 2023 to June 2024. Of the 2809 institutions, the University of California System emerged as the leading institution in ChatGPT research, contributing 142 articles, which accounted for 4.4% of the total number of articles. Harvard University followed with 113 articles (3.5%), and the State University System of Florida ranked third with 96 articles (3.0%). Overall, the countries' distribution showed the leading role of American institutions in advancing ChatGPT research.

Table 2. Top 10 productive institutions in ChatGPT research indexed by the Web of Science from January 2023 to June 2024.

Rank	Institution (Country)	Number of Articles (%)	% of 11,357 Institutions
1	University of California System (USA)	142 (4.4)	1.25
2	Harvard University (USA)	113 (3.5)	0.99
3	State University System of Florida (USA)	96 (3.0)	0.84
4	National University of Singapore (Singapore)	85 (2.6)	0.75
5	Chinese Academy of Sciences (China)	81 (2.5)	0.71
6	Stanford University (USA)	74 (2.3)	0.65
7	Vanderbilt University (USA)	68 (2.1)	0.60
8	King Saud University (Saudi Arabia)	65 (2.0)	0.57
9	University System of Ohio (USA)	65 (2.0)	0.57
10	University of Toronto (Canada)	60 (1.8)	0.53

3.4. Distribution of Journals and Disciplines

Table 3 lists the top 10 journals published original articles on ChatGPT indexed by the Web of Science from January 2023 to June 2024. The *Cureus Journal of Medical Science* published the highest number of original articles on ChatGPT, with 189 articles, representing 5.8% of the 3231 original articles. Despite its large output, this journal held a modest 2023 Journal Impact Factor of 1.0 and was ranked in the third quartile of the general and internal medicine category of Emerging Sources Citation Index (ESCI). *JMIR Medical Education*, with a higher impact factor of 3.2, followed with 50 articles, ranking in the first quarter in the scientific disciplines of education category. *IEEE Access* published 43 articles and held an impact factor of 3.4, positioned in Q2 for both computer science and engineering categories. Overall, the distribution of the journals reflected a diverse and multidisciplinary approach to ChatGPT research.

Table 3. The top 10 journals published original articles on ChatGPT indexed by the Web of Science from January 2023 to June 2024.

Rank	Journal (Publisher)	Number of Articles (%)	2023 Journal Impact Factor (Editon)	Web of Science Category [Journal Impact Factor Quartile]
1	*Cureus Journal of Medical Science* (Springer Nature)	189	1.0 (ESCI)	Medicine, General, and Internal [Q3]
2	*JMIR Medical Education* (JMIR Publications)	50	3.2 (ESCI)	Education, Scientific Disciplines [Q1]
3	*IEEE Access* (Institute of Electrical Electronics Engineers)	43	3.4 (SCIE)	Computer Science, Information Systems [Q2] Engineering, Electrical, and Electronic [Q2]
4	*Journal of Medical Internet Research* (JMIR Publications)	39	5.8 (SCIE)	Health Care Sciences and Services [Q1] Medical Informatics [Q1]

Table 3. *Cont.*

Rank	Journal (Publisher)	Number of Articles (%)	2023 Journal Impact Factor (Editon)	Web of Science Category [Journal Impact Factor Quartile]
5	*Scientific Reports* (Nature Portfolio)	37	4.6 (SCIE)	Multidisciplinary Sciences [2]
6	*Education and Information Technologies* (Springer)	34	4.8 (SSCI)	Education and Educational Research [Q1]
7	*European Archives of Oto-Rhino-Laryngology* (Springer)	28	1.9 (SCIE)	Otorhinolaryngology [Q2]
8	*Applied Sciences-Basel* (MDPI)	25	2.5 (SCIE)	Chemistry, Multidisciplinary [Q2] Engineering, Multidisciplinary [Q1] Materials Science, Multidisciplinary [Q3] Physics, Applied [Q2]
9	*International Journal of Human-Computer Interaction* (Taylor & Francis)	24	3.4 (SSCI and SCIE)	Computer Science, Cybernetics [Q2] Ergonomics [Q1]
10	*Journal of Chemical Education* (American Chemical Society)	22	2.5 (SCIE)	Chemistry, Multidisciplinary [Q2] Education, Scientific Disciplines [Q2]

ESCI: Emerging Sources Citation Index; SCIE: Science Citation Index Expanded; SSCI: Social Sciences Citation Index.

3.5. Distribution of Prolific Authors

Table 4 shows the top 10 most prolific authors with the highest number of original articles on ChatGPT indexed by the Web of Science from January 2023 to June 2024. The top three most prolific authors were Fei-Yue Wang, Jerome R. Lechien, and Ishith Seth. Fei-Yue Wang led with 28 original articles and boasted a substantial h-index of 67, derived from his entire body of publications. Jerome R. Lechien and Ishith Seth both contributed 16 articles each. Lechien had an h-index of 35, reflecting his significant impact and established career in research across all his publications, while Seth, with an h-index of 11, indicated his emerging prominence in the field.

Table 4. The top 10 prolific authors with highest number of original articles on ChatGPT indexed by the Web of Science from January 2023 to June 2024.

Rank	Name of Author	Number of Original Articles (%)	H-Index [a]
1	Wang, Fei-Yue	28	67
2	Lechien, Jerome R.	16	35
3	Seth, Ishith	16	11
4	Cheungpasitporn, Wisit	13	36
5	Mondal, Himel	13	11
6	Thongprayoon, Charat	13	37
7	Miao, Jing	12	45
8	Sohail, Shahab Saquib	12	8
9	Madsen, Dag Oivind	11	11
10	Rozen, Warren M.	11	42

[a] H-index was obtained from the Web of Science on 1 July 2024.

3.6. Citation Analysis

Citation analysis revealed the impact and influence of ChatGPT-related research. The total number of citations for ChatGPT publications from January 2023 to June 2024 was 3231, with an average of 5.6 citations per article. Table 5 lists the top 10 original articles on

ChatGPT research with the highest citations indexed by the Web of Science from January 2023 to June 2024. The leading article, authored by Dwivedi et al. and titled "Opinion paper: 'So what if ChatGPT wrote it?' Multidisciplinary perspectives on opportunities, challenges and implications of generative conversational AI for research, practice and policy" published in the *International Journal of Information Management*, received 564 citations. The second most cited article, by Ayers et al., compared physician and AI chatbot responses to patient questions on a public social media forum, published in *JAMA Internal Medicine* with 400 citations. The third top-cited work by Alkaissi and McFarlane, titled "Artificial hallucinations in ChatGPT: Implications in scientific writing" appeared in the *Cureus Journal of Medical Science* and received 312 citations. Together, these highly cited papers reflected the academic and practical interest in understanding and harnessing ChatGPT's capabilities and implications.

Table 5. The top 10 original articles on ChatGPT research with highest citations indexed by the Web of Science from January 2023 to June 2024.

Rank	First Author (No. of Total Authors)	Article Title (DOI)	Journal	Total Citations
1	Dwivedi YK (77)	Opinion paper: "So what if ChatGPT wrote it?" Multidisciplinary perspectives on opportunities, challenges and implications of generative conversational AI for research, practice and policy (https://doi.org/10.1016/j.ijinfomgt.2023.102642)	*International Journal of Information Management*	564
2	Ayers JW (3)	Comparing physician and artificial intelligence chatbot responses to patient questions posted to a public social media forum (https://doi.org/10.1001/jamainternmed.2023.1838)	*JAMA Internal Medicine*	400
3	Alkaissi H (2)	Artificial hallucinations in ChatGPT: Implications in scientific writing (https://doi.org/10.7759/cureus.35179)	*Cureus Journal of Medical Science*	312
4	Cotton DRE (3)	Chatting and cheating: Ensuring academic integrity in the era of ChatGPT (https://doi.org/10.1080/14703297.2023.2190148)	*Innovations in Education and Teaching International*	252
5	Tlili A (7)	What if the devil is my guardian angel: ChatGPT as a case study of using chatbots in education (https://doi.org/10.1186/s40561-023-00237-x)	*Smart Learning Environments*	249
6	Cascella M (4)	Evaluating the feasibility of ChatGPT in healthcare: An analysis of multiple clinical and research scenarios (https://doi.org/10.1007/s10916-023-01925-4)	*Journal of Medical Systems*	235
7	Salvagno M (3)	Can artificial intelligence help for scientific writing? (https://doi.org/10.1186/s13054-023-04380-2)	*Critical Care*	191
8	Khan RA (4)	ChatGPT—Reshaping medical education and clinical management (https://doi.org/10.12669/pjms.39.2.7653)	*Pakistan Journal of Medical Sciences*	164
9	Farrokhnia M (4)	A SWOT analysis of ChatGPT: Implications for educational practice and research (https://doi.org/10.1080/14703297.2023.2195846)	*Innovations in Education and Teaching International*	157
10	Cooper G (1)	Examining science education in ChatGPT: An exploratory study of generative artificial intelligence (https://doi.org/10.1007/s10956-023-10039-y)	*Journal of Science Education and Technology*	142

DOI: Digital Object Identifier.

3.7. Keyword Analysis

Figure 2 shows a word cloud of the 100 most prominent author's keywords of original articles on ChatGPT research indexed by the Web of Science from January 2023 to June 2024. As expected, the most prominent terms were "artificial intelligence", "ChatGPT", "large language models", and "natural language processing", reflecting the central topics of interest. Key, related terms included "generative AI", "machine learning", and "chatbots", indicating a strong emphasis on the development and application of conversational agents. In addition, the prominence of words such as "education", "medical education", "patient education", and "higher education" suggested a focus on the impact of ChatGPT in educational settings. Other notable terms such as "ethics", and "academic integrity" showed the importance of ethical considerations. The presence of technical terms such as "prompt engineering", "deep learning", and "transformers" reflected the ongoing research into the ChatGPT technology. Overall, the word cloud included a broad and multidisciplinary engagement with ChatGPT including technological, educational, ethical, and healthcare domains.



Figure 2. Word cloud of the 100 most prominent authors' keywords based on square root of word occurrence of original articles on ChatGPT research indexed by the Web of Science from January 2023 to June 2024.

Figure 3 shows a thematic mapping based on a Louvain clustering algorithm of 250 authors' keywords of original articles on ChatGPT research indexed by the Web of Science from January 2023 to June 2024. The analysis indicated several key themes in ChatGPT research represented by clusters of themes categorized by their centrality and density. In the niche themes quadrant, topics like "writing", "social media", and "communication" are prominent, indicating their specialized and developed nature. Basic themes included "ChatGPT", "artificial intelligence", and "generative AI", reflecting their widespread but fundamental role in the research landscape. The central cluster, comprising "natural language processing", "chatbot", and "medical education", indicated these themes are both developed and central to current research. The motor themes quadrant is empty, indicating a lack of highly developed and influential themes in the dataset. Emerging or declining themes, represented by the lower left quadrant, are also devoid of clusters, suggesting no themes in this area are currently considered underdeveloped or losing relevance. In addition, a distinct cluster in the mid-range of density and centrality includes "chatbots", "deep learning", and "task analysis", suggesting these as significant but not yet fully mature research areas.

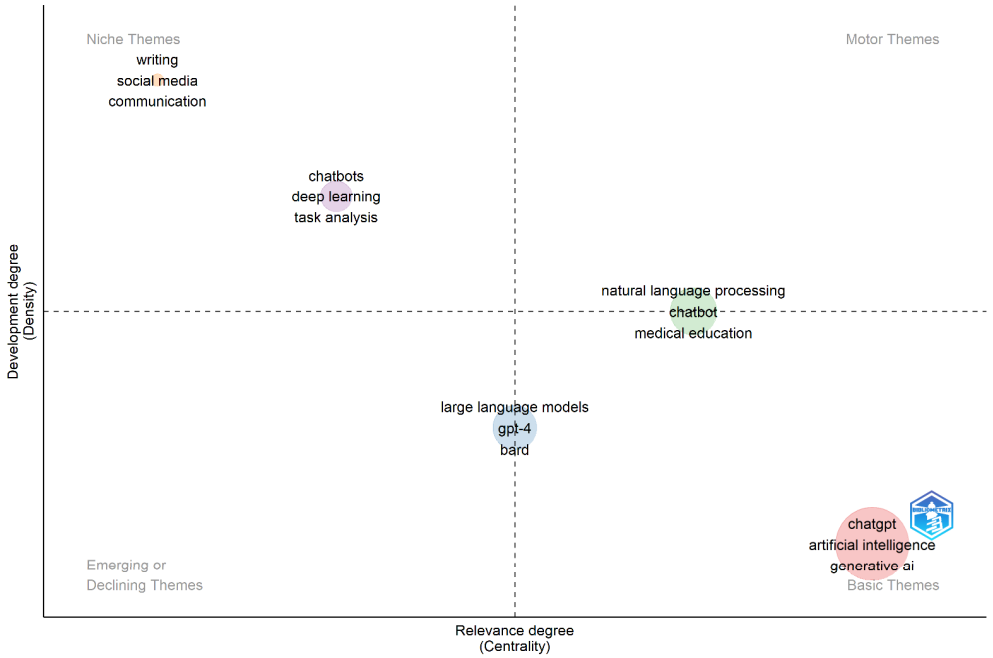


Figure 3. A thematic mapping of authors' keywords of original articles on ChatGPT research indexed by the Web of Science from January 2023 to June 2024.

3.8. Keyword Co-Occurrence Network

Based on the keywords co-occurrence network analysis, four interconnected clusters were revealed, showing the interdisciplinary nature of ChatGPT's research in technology, healthcare, and education (Figure 4). Specifically, cluster 1 (red) contained key terms such as "acceptance", "adoption", "attitude", "communication", and "knowledge". It primarily focused on how individuals and organizations responded to and integrated new technologies, particularly in education and healthcare. Cluster 2 (green) focused on AI and machine learning, with prominent keywords including "accuracy", "AI chatbot", "artificial intelligence", "machine learning", and "natural language processing". The cluster explored the development, application, and impact of AI and machine learning technologies across healthcare and education. Cluster 3 (blue) centered on academic integrity, writing, and research methodologies. Key terms in this cluster were "academic integrity", "academic writing", "authorship", "plagiarism", and "AI ethics". This cluster emphasized the importance of maintaining ethical standards in research and addressed the challenges associated with academic writing and publication. Cluster 4 (yellow) focused on the information provided by AI. Important keywords include "patient education", "literacy information", "quality", and "internet". This cluster showed the role of ChatGPT in healthcare outcomes and patient education.

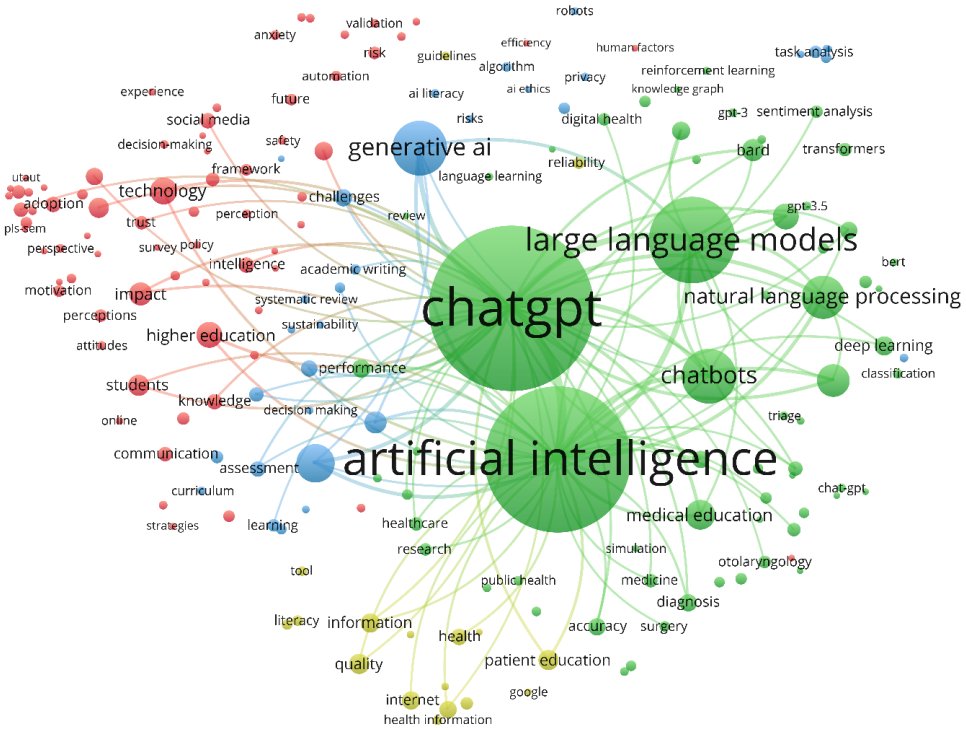


Figure 4. A co-occurrence network analysis of authors' keywords and Keywords Plus of original articles on ChatGPT research indexed by the Web of Science from January 2023 to June 2024.

4. Discussion

This bibliometric analysis provides a comprehensive overview of ChatGPT-related research from January 2023 to June 2024. The findings showed the substantial growth and widespread interest in ChatGPT across various disciplines, particularly in healthcare, education, and technology.

The annual growth rate of 37.6% in ChatGPT-related publications demonstrated an accelerating interest in this technology. The average citation rate of 5.6 citations per article indicated that these works had received attention and contributed to ongoing discussions. The analysis of authorship patterns revealed a high level of collaboration, with 26.7% of articles involving international co-authorship. This reflected the global interest in and collaborative nature of research on ChatGPT, as researchers across the world are keen to explore its potential and implications.

The United States of America emerged as the leading country in terms of publication output, followed by China, India, and Germany. The high MCP percentages for China and Germany suggested robust international collaboration, which may enhance the quality and impact of research through diverse perspectives and expertise. Leading institutions such as the University of California System, Harvard University, and the State University System of Florida were at the forefront of this research.

The distribution of journals and disciplines revealed the multidisciplinary nature of ChatGPT research. Journals such as the *Cureus Journal of Medical Science*, *JMIR Medical Education*, and *IEEE Access* reflected the diverse applications of ChatGPT in medical science, education, and engineering. This diversity showed the broad impact of ChatGPT across different fields and its potential to drive innovation in various sectors.

The three top cited articles, by Dwivedi et al. [32], Ayers et al. [33], and Alkaissi and McFarlane [34], share a common focus on the implications and challenges of generative AI, particularly in the context of healthcare and scientific writing. The high citation of these articles reflects the critical importance of trust in AI outputs, whether in healthcare

communications or academic writing. The potential for misinformation or "hallucinations" from generative AI systems raises ethical concerns about their use in sensitive areas like health and research. Thus, there is a pressing need for frameworks that address the ethical and practical challenges identified in these studies.

Keyword analysis and thematic mapping revealed several key themes in ChatGPT research. The broad and multidisciplinary engagement seen in the word cloud reflected the diverse applications and implications of ChatGPT technology across various sectors. Central topics included AI, large language models, natural language processing, and generative AI, reflecting the core technological focus. Prominent terms, such as "medical education" and "patient education", indicated an interest in applying ChatGPT in educational settings. Ethical considerations, evidenced by keywords like "ethics" and "academic integrity", were also prominent, showing the importance of addressing ethical challenges associated with AI technologies.

The thematic map analysis revealed several insights into the current research trends and potential future directions in ChatGPT research. The center cluster's emphasis on "natural language processing" and "chatbot" showed their importance and extensive development within the research community. This suggested a strong focus on improving chatbot functionalities and their applications across various domains, including "medical education". The presence of "large language models", "GPT-4", and "Bard" in the lower central area indicated these advanced AI systems are still under development. The lack of clusters in the motor quadrant suggested a potential gap in highly influential and rapidly advancing research areas. The absence of themes in the emerging or declining quadrant suggested a stable research landscape without major shifts, likely because ChatGPT research is still in its early stages. Furthermore, the presence of "writing", "social media", and "communication" in the niche themes quadrant reflected specialized yet well-developed areas. The mid-range cluster of "chatbots", "deep learning", and "task analysis" showed growing interest and development in these areas.

The keyword co-occurrence network further illustrated the interdisciplinary nature of ChatGPT research. The network revealed four primary clusters, each representing distinct but interconnected areas of focus: technology adoption, AI development, academic integrity, and healthcare applications. This clustering illustrated the breadth and depth of ChatGPT research and highlights the multifaceted impact of this technology. Cluster 1, focused on technology adoption, emphasized the social and organizational dimensions of integrating ChatGPT into various sectors. It suggested a significant interest in understanding how individuals and organizations respond to and adopt new AI technologies. Cluster 2 concentrated on AI development, reflecting the technical advancements and methodological innovations driving ChatGPT research. The focus on accuracy and natural language processing reflected the ongoing efforts to refine AI capabilities and improve interaction quality. Cluster 3 was centered on academic integrity, writing, and research methodologies, illustrating the ethical considerations and challenges associated with AI in academic contexts. The emphasis on integrity and ethics is particularly relevant as ChatGPT's capabilities raise concerns about plagiarism [35], authorship [36], and the responsible use of AI-generated content [37–39]. As ChatGPT technology continues to evolve, it is crucial to monitor its development and applications to understand its long-term implications and benefits. Finally, cluster 4 addressed the role of ChatGPT in patient education and healthcare information, showing its impact on literacy and the quality of information available to patients. These clusters illustrated the diverse applications and challenges associated with ChatGPT, including technological development, ethical considerations, and practical applications in education and healthcare.

4.1. Study Limitations

This bibliometric analysis has several limitations. Firstly, the analysis was limited to publications indexed in the Web of Science, which may not capture all the relevant literature, particularly from less prominent journals. This could result in an incomplete picture of the global research landscape. Future studies could use data from other bibliometric databases, such as Scopus, PubMed, and Google Scholar. Nevertheless, not all databases provide citation data, which limits unified citation analysis across multiple databases. For instance, PubMed offers extensive article coverage but lacks the comprehensive citation tracking found in WoS. Secondly, the citation metrics used to assess impact and influence might not fully reflect the quality and practical significance of the research. In addition, the rapid evolution of AI technologies means that the findings may quickly become outdated, necessitating continuous updates to maintain relevance.

4.2. Study Implications

The findings of this bibliometric analysis have several implications for future research. The rapid growth and widespread interest in ChatGPT suggested that it will continue to be a significant area of study. As AI becomes more integrated into healthcare and research, it is crucial to address concerns related to data privacy, algorithmic bias, and the misuse of AI-generated content. There is a pressing need for the development of comprehensive policies and regulations that govern the responsible use of AI in biomedical research. These guidelines should be collaboratively developed by a diverse group of stakeholders, including researchers, healthcare professionals, ethicists, policymakers, and representatives from patient advocacy groups. Such collaboration will ensure that the policies are well-rounded, taking into account the perspectives and needs of all the parties involved.

It should also be mentioned that robust evaluation metrics are essential to assess the reliability and safety of large language models (LLMs) such as ChatGPT. The recent research has highlighted the challenges of evaluating LLMs, emphasizing the importance of reproducibility, reliability, and robustness. Factors such as prompt engineering, decoding parameters, and data integrity issues (e.g., contamination and outdated labels) significantly impact evaluation outcomes [40]. In healthcare, where data analytics is critical, integrating LLMs requires careful assessment of their capabilities [41]. Furthermore, the rise of multi-modal LLMs necessitates extending evaluation methodologies to include visual and linguistic processing [42]. Incorporating these advancements will enable rigorous analysis, support responsible clinical implementation, and mitigate risks such as bias, hallucinations, and inaccuracies.

The significant focus on medical education and patient education indicates a growing interest in using AI technologies to enhance educational outcomes. AI-driven tools, such as personalized learning platforms, virtual tutors, and interactive simulations, have the potential to revolutionize medical training by providing customized learning experiences, improving accessibility, and enhancing the retention of complex concepts. Therefore, it is essential to develop and implement policies that ensure these technologies are used effectively and ethically. Accreditation bodies should collaborate with AI developers, educators, and healthcare professionals to create comprehensive guidelines that ensure AI tools contribute positively to medical education. Moreover, policies should encourage continuous professional development for educators to ensure they are well-equipped to incorporate AI into their teaching practices. In addition, guidelines for the ethical use of AI should be embedded within the educational curriculum. This includes not only teaching students about the ethical implications of AI but also ensuring they understand the importance of data privacy and responsible data usage [43].

5. Conclusions

This bibliometric analysis provides an overview of the collaboration patterns and key research themes in ChatGPT-related studies from January 2023 to June 2024. The findings showed the multidisciplinary nature of this field, revealing significant intersections across technological innovation, healthcare applications, educational advancement, and ethical considerations. Several insights emerged regarding current research trends and potential future directions, including enhancing chatbot functionalities, advancing AI system capabilities, and expanding applications in deep learning and task analysis. The analysis also emphasizes the importance of ethical considerations in chatbot technology development to ensure a sustainable and positive societal impact.

This analysis addresses key research gaps by synthesizing global research trends and validating the relevance of advanced AI applications, while also identifying areas requiring further methodological refinement. The findings support the responsible integration of ChatGPT into patient education and clinical decision-making and inform the development of AI-enhanced learning platforms, particularly for transforming medical training and enabling personalized education. These insights contribute to a deeper understanding of ChatGPT's evolving role across multiple disciplines and establish a foundation for future research.

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